

Local-Unitary Rigidity of Stabilizer AME States and Transversal Clifford Groups of MDS–CSS Codes

Tavis Rudd

July 2026

Abstract

Let $q = p^e$. For every $m \geq 2$, each product-unitary intertwiner between two stabilizer $\text{AME}(2m, q)$ states is Clifford factor by factor. On any $m + 1$ parties, the supported stabilizer labels form a q^2 -element group and project bijectively onto the complete local Weyl basis at every party in the marginal. The reduced operator therefore determines its local Weyl axes intrinsically. Consequently, every tensor-product conversion between the associated $[[2m - 1, 1, m]]_q$ stabilizer quantum-MDS encoders is Clifford on every physical factor and on the logical factor. Modulo party relabelling and local trace-symplectic $\mathbb{F}_p$ -linear changes of frame, the transition atlas obtained by pushing Weyl labels between parties on minimum supports is a complete structured LU invariant.

For equal-phase CSS states of linear $[2m, m, m + 1]_q$ MDS codes over an odd prime field, the exact fixed-party projective transversal logical group is $\mathbb{F}_q^2 \rtimes \text{SL}_2(q)$ precisely when C is diagonally isodual, and $\mathbb{F}_q^2 \rtimes T$ otherwise, where T is the split torus; generalized Reed–Solomon codes lie on the first branch.

For $m = 3$, a degree-eight quotient z classifies projective and monomial-code equivalence on an explicit non-GRS pencil of six-point arcs over odd fields; over odd prime fields it also classifies local-Clifford and local-unitary equivalence.

1 Introduction

An absolutely maximally entangled state has maximally mixed reductions on every subsystem no larger than half the parties. Fix a prime power $q = p^e$ and the standard additive q -dimensional Weyl system. We ask whether a stabilizer $\text{AME}(2m, q)$ state retains its local Weyl frames under arbitrary single-party changes of basis.

The assertion that local-unitary (LU) equivalence implies local-Clifford (LC) equivalence fails for general stabilizer states [JCWY10]. Maximal entanglement supplies the missing rigidity.

Throughout, *transversal* means a tensor product of one-site physical operators with the party labels fixed. A *fixed-party* symmetry has this form; a *party-moving* symmetry additionally permutes tensor factors. LU and LC equivalence allow a party permutation when it is displayed explicitly. Logical groups called fixed-party below exclude such permutations.

Theorem 1.1 (Local-unitary rigidity). Let $q = p^e$, let $m \geq 2$, and let $|\psi\rangle, |\phi\rangle$ be stabilizer $\text{AME}(2m, q)$ states for the standard additive Weyl system. Suppose that, for a party permutation π , single-qudit unitaries $U_1, \dots, U_{2m}$, and a phase θ ,

$$(U_1 \otimes \cdots \otimes U_{2m})\pi|\psi\rangle = e^{i\theta}|\phi\rangle.$$

Then every U_i normalizes the finite-field Weyl system. In particular, the displayed equivalence is local Clifford.

The mechanism is visible in any $m+1$ -party reduced operator. Stabilizer labels supported on those parties form a group of order q^2 , and projection to the local Pauli-label group at every party in the marginal is bijective. The marginal is therefore diagonal on the complete local Weyl basis, with nonzero coefficients. A rank-one contraction recovers exactly one Weyl axis, so the tensor intrinsically determines every local Weyl frame. Product conjugation must permute those frames and is consequently Clifford. The argument uses neither CSS form, equal phases, classical linearity, nor $\mathbb{F}_q$ -linearity of the stabilizer labels. Thus the proof has four steps:

$$\begin{aligned} \text{AME support squeeze} &\implies \text{full-Weyl marginal} \\ &\implies \text{intrinsic local axes} \implies \text{local Clifford factors.} \end{aligned}$$

Corollary 1.2 (Transversal Clifford no-go). Let $V_\psi, V_\phi : \mathbb{C}^q \rightarrow (\mathbb{C}^q)^{\otimes(2m-1)}$ be encoding isometries whose normalized Choi states are stabilizer AME($2m, q$) states $|\psi\rangle, |\phi\rangle$. They encode the associated $[[2m-1, 1, m]]_q$ quantum MDS codes. If

$$(U_1 \otimes \cdots \otimes U_{2m-1})V_\psi = V_\phi L$$

for a logical unitary L , then L and every U_i are Clifford. Thus every transversal conversion between two codes in the family is Clifford factor by factor. Taking $V_\psi = V_\phi$, no such code admits a transversal non-Clifford logical unitary.

The support bijections also retain how local Weyl frames on different parties fit together. On an $(m+1)$ -party support A , let $p_i : L_\psi(A) \rightarrow \mathbb{F}_q^2$ be the bijective projection at party i . The maps $M_A(i, j) = p_j p_i^{-1}$ push a Weyl label from i to j . Their collection over all A, i, j is the *minimum-support atlas*. Local trace-symplectic frame changes are compatible precisely when they intertwine every $M_A(i, j)$. Here a projective product-unitary group means the image of the product-unitary symmetries in the projective unitary group of the full tensor product. Schematically, overlapping supports turn operator pushing into a loop,

$$P_i \xrightarrow{M_A(i, j)} P_j \xrightarrow{M_B(j, k)} P_k \xrightarrow{M_C(k, i)} P_i,$$

whose composite is an atlas holonomy.

Theorem 1.3 (Minimum-support atlas classification). After a fixed party relabelling, two stabilizer AME($2m, q$) states are LU equivalent exactly when their minimum-support operator-pushing atlases are equivalent by local symplectic changes of Weyl frame. For one state, let $\mathcal{G}_\psi$ be its group of compatible local symplectic gauges, let Γ_ψ be its fixed-party projective product-unitary symmetry group, and identify its projective Pauli stabilizer with the label group L_ψ . Then

$$1 \longrightarrow L_\psi \longrightarrow \Gamma_\psi \longrightarrow \mathcal{G}_\psi \longrightarrow 1$$

is exact. If q is prime, choosing base transports identifies $\mathcal{G}_\psi$ with the centralizer in $\text{SL}_2(q)$ of the atlas holonomies.

Thus an a priori continuous LU-equivalence problem reduces to finite symplectic linear algebra on Weyl frames recovered from the state itself.

Rains recovered the three Pauli axes from a diagonal correlation tensor, and Van den Nest, Dehaene, and De Moor used this mechanism in a minimal-support LU-to-LC criterion for qubit stabilizer states [Rai99, Theorem 13] [VdNDDM05b, Theorem 1 and Corollary 1]. For $q = 2$, Proposition 3.2 supplies all three Pauli axes on each minimum support, so the AME profile satisfies their criterion. The LU-to-LC conclusion of Theorem 1.1 is therefore already covered by their result. The new scope here is the uniform prime-power theorem for arbitrary additive stabilizers, together with the minimum-support atlas and its exact symmetry-group sequence. Van den Nest, Dehaene, and De Moor also gave a finite invariant set characterizing

LC equivalence of arbitrary binary stabilizer states [VdNDDM05a]. Our atlas theorem is not a first general LC-classification result: it uses the AME support profile to replace a global invariant list by an all-prime-power operator-pushing criterion and an explicit symmetry-group centralizer. Tan relates the LU orbits and local symmetries of even-party AME tensors to quantum-MDS codes and computes the full local symmetry group of the canonical four-qutrit AME state [Tan26, Theorems 3.5, 3.6, and 5.3]. The local factors in Tan’s displayed generators are qutrit Cliffords, so that calculation contains the $q = 3, m = 2$ automorphism subcase above. Our contributions are the uniform all-prime-power, all- m , arbitrary-additive intertwiner theorem and the exact minimum-support-atlas classification of its residual Clifford symmetries. The AME condition itself supplies both the full-Weyl marginal required by the axis argument and the minimum-support generation required by the atlas theorem.

Equivalently, the support bijection is a minimal-support Pauli form of perfect-tensor operator pushing: every one-party Weyl operator has a unique product-Weyl representative on any chosen m of the other parties. Perfect-tensor operator pushing and the associated $[[2m - 1, 1, m]]$ encoding interpretation were introduced in the holographic-code setting by Pastawski, Yoshida, Harlow, and Preskill [PYHP15, Eq. (2.3) and Section 5.7]. Overlapping minimum supports compare the recovered local Weyl-label frames. Theorem 1.3 makes their transition atlas a complete classification reduction rather than only an obstruction. For six parties, explicit holonomy conjugacy data recover the pencil coordinate z . The boundary $m \geq 2$ is sharp: a Bell pair has non-Clifford $U \otimes \bar{U}$ symmetries.

Rigidity determines which product operations are possible; the maximum-distance-separable (MDS) Calderbank–Shor–Steane (CSS) model determines which Clifford operations actually occur. If $C \leq \mathbb{F}_q^{2m}$ is a linear $[2m, m, m + 1]_q$ MDS code, then

$$|\Psi_C\rangle = q^{-m/2} \sum_{c \in C} |c\rangle \quad (1.1)$$

is a stabilizer $\text{AME}(2m, q)$ state. Conversely, an equal-phase state of a linear $[2m, m]_q$ code is AME only if the code is MDS [RGRA18]. Here the MDS–CSS geometry is no longer needed for rigidity; it makes the exact fixed-party projective logical image computable.

Corollary 1.4 (Diagonal-isodual transversal logical group). Let q be an odd prime, let $m \geq 2$, and let $C \leq \mathbb{F}_q^{2m}$ be a linear $[2m, m, m + 1]_q$ MDS code. The projective group of logical unitaries of the associated $[[2m - 1, 1, m]]_q$ quantum MDS code that admit tensor-product physical implementations is

$$\begin{cases} \mathbb{F}_q^2 \rtimes \text{SL}_2(q), & \text{if } SC = C^\perp \text{ for a nonsingular diagonal } S, \\ \mathbb{F}_q^2 \rtimes T, & \text{otherwise,} \end{cases}$$

where $T = \{\text{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}$.

Put

$$\mathcal{D}(C, C^\perp) = \{s \in \mathbb{F}_q^{2m} : \text{diag}(s)C \subseteq C^\perp\}.$$

The branch condition is the nullity test $\dim \mathcal{D}(C, C^\perp) = 1$. Proposition 3.8 proves that this dimension is zero or one and that every nonzero vector in the space has full support and is unique up to scalar.

The nullity-one condition governs the two branches above. In odd characteristic the $\text{SL}_2(q)$ factor has coherent Weil lifts, while the affine projective group retains the usual Heisenberg central extension; Section 3 separates these lift questions. Generalized and extended generalized Reed–Solomon codes are diagonally isodual, but the condition is not confined to that family: self-dual non-GRS MDS constructions are known [ZL24].

Table 1: Scopes of the principal results. “Fixed after relabelling” means that a displayed party permutation is applied first and the remaining equivalence is fixed-party.

Result	Local dimension	State or code assumptions	Party motion
LU rigidity	p^e	arbitrary stabilizer AME($2m, q$)	allowed explicitly
Atlas classification	p^e	arbitrary stabilizer AME($2m, q$)	fixed after relabelling
Exact fixed-party projective logical group	odd prime	equal-phase linear [$2m, m, m + 1$] MDS–CSS	fixed
Pencil LU/LC by z	odd prime	admitted non-GRS six-party pencil	arbitrary permutation

For example, an extended $[8, 4, 5]_7$ Reed–Solomon code gives an AME($8, 7$) tensor and a $[[7, 1, 4]]_7$ quantum MDS code whose projective transversal logical group has order $7^2|\mathrm{SL}_2(7)| = 16464$. No non-Clifford transversal gate enlarges this group.

For binary stabilizer codes, Wirthmüller determines the identity component of the local-unitary automorphism group and gives a protected-qubit criterion for finiteness of the induced automorphism group [Wir11, Theorem 7 and Corollary 11]. Our result instead determines every local factor and gives projective finiteness uniformly inside the prime-power stabilizer-AME class. The stabilizer qualifier is essential: arbitrary phased minimal-support AME($6, d$) states form infinitely many LU classes [RL25]; nonlinear orthogonal arrays and nonstabilizer AME states are not covered.

The geometric applications in the remainder of the paper specialize to $m = 3$. Our main explicit application is the ordered pencil

$$H_t = ((0, 1, 1 - t), (0, 1, t - 1), (1, 1 - t, 0), (1, t - 1, 0), (1, 0, -t), (1, 0, t)). \quad (1.2)$$

On its admitted odd-field non-GRS locus, define

$$\begin{aligned} A(t) &= -4t(t - 1)^2, \\ B(t) &= (t^2 - t + 1)(t^2 - 3t + 1), \\ z(t) &= (B(t)/A(t))^2. \end{aligned} \quad (1.3)$$

Exact bracket identities show that z is the degree-eight projective quotient of the pencil. Theorem 1.1 upgrades its local-Clifford classification to the following complete local result.

Corollary 1.5 (Classification of the admitted pencil). Let t, u lie in the nondegenerate locus (4.1) over an odd prime field $\mathbb{F}_q$. Write $\sim_{\mathrm{proj}}$ for projective equivalence of the ordered six-arcs up to a party permutation and $\sim_{\mathrm{mon}}$ for monomial equivalence of their codes. Then

$$H_t \sim_{\mathrm{proj}} H_u \iff \ker H_t \sim_{\mathrm{mon}} \ker H_u \iff \Psi_t \sim_{\mathrm{LC}} \Psi_u \iff \Psi_t \sim_{\mathrm{LU}} \Psi_u \iff z(t) = z(u),$$

where party permutations are allowed.

The exact diagonal-isodual group dichotomy above holds for every m . For six parties it has a geometric coding-theoretic interpretation over odd prime fields. Regard any one tensor leg as the input of the associated $[[5, 1, 3]]_q$ encoder. The fixed-party product-Clifford symmetries induce the full logical $\mathrm{SL}_2(q)$ precisely when the six-arc is conic/GRS; otherwise they induce only the split torus. Thus the nineteenth-century self-association condition for six points controls which one-qudit logical Clifford gates are available without moving parties. Including logical Paulis, the exact fixed-party projective logical group is $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ on the GRS locus and $\mathbb{F}_q^2 \rtimes T$ off it.

Quantum Reed–Solomon and polynomial-code constructions provide the standard coding setting [GGB99, ABO08, Got99]. Closest to our fixed-party question are restrictions on diagonal gates between qubit codes [AJO14, Propositions 3 and 4], the endomorphism-algebra classification of multiblock diagonal Clifford gates [DB25, Theorems 5.5 and 6.1], and automorphism methods that recover SWAP-transversal logical Cliffords and their Pauli corrections [SKW⁺25]. Other locality models, including fold-transversal and 2-local layers, can realize larger groups [TCT26, Hol26]; fully transversal gates remain subject to the Eastin–Knill obstruction and, for codes with several logical qubits, cannot realize the full logical Clifford group [EK09, CG26].

Our coordinates may carry independent one-qudit factors, and Corollary 1.2 also treats conversions between different one-logical-qudit quantum-MDS encoders. The MDS–CSS specialization identifies the exact fixed-party projective image for every half-dimensional MDS code over an odd prime field. Its dividing condition, diagonal code–dual equivalence, also occurs beyond GRS codes in algebraic-geometry MDS families [ZZ26].

Classically, we use Coble’s association theory [Cob22] through the modern treatment of Dolgachev and Ortland [DO88, Chapter III]; for finite-projective arc background see Segre, Hirschfeld, and the modern survey of Ball and Lavrauw [Seg55, Hir98, BL19]. Over characteristic zero, the GIT quotient of six ordered points in $\mathbb{P}^2$ is a double cover of $\mathbb{P}^4$, with Gale association exchanging the sheets and the self-associated locus forming the Igusa-quartic branch divisor [HMSV08, Section 2.3]. That divisor is dual to the Segre cubic parametrizing six points on $\mathbb{P}^1$. Our six-arc phase theorem is the finite-field, fixed-label slice of this picture: the branch condition becomes diagonal isoduality, and the Veronese determinant detects the conic/GRS locus. This interpretation is not used in the proof.

Sections 2 and 3 contain the main line: the arc–code–AME conventions, LU-to-LC rigidity, and its structural completion by the minimum-support atlas. The next three sections develop applications and contrasts. Section 4 proves the z -classification; Section 5 proves the logical-Clifford phase theorem; and Section 6 compares operator-valued rigidity with the generic blindness of fixed-copy scalar invariants. Section 7 records the trust boundary, and Section 8 closes with the exact extension-field boundary. The appendices contain the detailed H_3 and $q = 13$ scalar certificates, optional transport calculations, and finite party-permutation examples; none is used in the uniform rigidity or atlas theorems.

2 Stabilizer AME states and MDS–CSS models

Let $q = p^e$ and fix the additive character $\chi(a) = \exp(2\pi i \operatorname{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(a)/p)$. On the standard basis of $\mathbb{C}^q$, indexed by $\mathbb{F}_q$, put

$$X(a)|x\rangle = |x + a\rangle, \quad Z(b)|x\rangle = \chi(bx)|x\rangle.$$

For $v = (a, b) \in \mathbb{F}_q^2$, write $W_v = X(a)Z(b)$, with any fixed consistent phase convention when only its projective axis is used. The q^2 single-party operators W_v form an orthogonal basis of the single-party Hilbert–Schmidt space. A unitary is Clifford when conjugation permutes their projective axes $\mathbb{C}W_v$.

After choosing dual $\mathbb{F}_p$ -bases of $\mathbb{F}_q$, this Weyl system is the Pauli system of e p -level factors. Its projective Clifford normalizer has the standard exact sequence

$$1 \longrightarrow \mathbb{F}_p^{2e} \longrightarrow \operatorname{PCliff}(q) \longrightarrow \operatorname{Sp}_{2e}(\mathbb{F}_p) \longrightarrow 1, \quad (2.0)$$

including $p = 2$: the kernel is the projective Pauli group, and every trace-symplectic $\mathbb{F}_p$ -linear label map has a Clifford lift [DDM03, HDDM05].

For n parties, a pure stabilizer state is fixed by an abelian Pauli group $\mathcal{S}$ of order q^n . Its projective label group $L \leq (\mathbb{F}_q^2)^n$, viewed additively over $\mathbb{F}_p$, is maximal isotropic for

the trace-symplectic form. This is the standard arbitrary-additive prime-power stabilizer correspondence [KKKS06, Theorems 13 and 15]. For each $\ell \in L$, write the stabilizer lift as $\widetilde{W}_\ell = \alpha_\ell W_\ell$, where $\alpha_\ell \neq 0$. Then

$$|\psi\rangle\langle\psi| = q^{-n} \sum_{\ell \in L} \widetilde{W}_\ell. \quad (2.1)$$

If A is a party set, put

$$L(A) = \{\ell \in L : \text{supp}(\ell) \subseteq A\}.$$

Weyl orthogonality and partial trace give

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L(A)} \widetilde{W}_\ell. \quad (2.2)$$

In particular, ρ_A^ψ is maximally mixed exactly when $L(A) = \{0\}$. Thus a stabilizer state on $2m$ parties is AME exactly when it has no nonidentity stabilizer label supported on at most m parties.

The rigidity and atlas theorems use only (2.0)–(2.2). For the geometric classification and explicit six-party applications, we now specialize to the MDS–CSS model.

A six-arc is an ordered sextuple of points of $\mathbb{P}^2(\mathbb{F}_q)$ with no three collinear. Choose nonzero column representatives and collect them in a 3-by-6 matrix H . The arc condition says that every three-column minor of H is nonzero. Consequently

$$C = \ker H \leq \mathbb{F}_q^6$$

is an $[6, 3, 4]_q$ MDS code, and its dual $C^\perp = \text{rowspan } H$ is MDS with the same parameters. Conversely, a parity-check matrix of such a code supplies a six-arc. Projectivities, independent rescaling of columns, and permutation of columns correspond exactly to monomial equivalences of the kernels. These objects exist exactly when $q \geq 4$. Indeed, from one point of a six-arc the five joining lines are distinct, so $q + 1 \geq 5$. For $q = 4$, a conic together with its nucleus is a six-arc; for $q \geq 5$, any six points of a nonsingular conic form a six-arc.

For the resulting CSS stabilizer we use the exact lift

$$W_{(c,h)} = X(c)Z(h) = \bigotimes_{i=1}^6 X(c_i)Z(h_i), \quad (c, h) \in C \oplus C^\perp.$$

Since $h \cdot c' = 0$ for $h \in C^\perp$ and $c' \in C$, these lifts multiply without a residual phase and fix $|\Psi_C\rangle$.

Proposition 2.1 (Arc–AME dictionary). For a six-arc with kernel C , the state $|\Psi_C\rangle$ in (1.1) is a minimal-support stabilizer $\text{AME}(6, q)$ state. Its projective Pauli-label Lagrangian is

$$L_C = (C \oplus 0) + (0 \oplus C^\perp) \leq \mathbb{F}_q^{12}.$$

Monomial code equivalence, together with a party permutation, induces local-Clifford equivalence of the corresponding states.

Proof. The operators $X(c)$, $c \in C$, fix the equal superposition. The operators $Z(h)$, $h \in C^\perp$, fix it because $\chi(h \cdot c) = 1$. The two subgroups commute, and their label space has dimension six and is Lagrangian.

For any three-coordinate set I , puncturing C to I is bijective: injectivity follows from distance four and both spaces have dimension three. In the partial trace of $|\Psi_C\rangle\langle\Psi_C|$, equality on the complementary three coordinates therefore forces equality of the codewords, while the

projection to I runs once through $\mathbb{F}_q^3$. Hence every three-party reduction is $q^{-3}I$. Smaller reductions are maximally mixed by tracing again, so the state is AME(6, q). Its support has the minimum possible size q^3 .

A column multiplier $x \mapsto ax$ is a single-party Clifford operation: it sends $X(b)$ to $X(ab)$ and $Z(b)$ to $Z(a^{-1}b)$. Projective row operations do not change the kernel, and column permutations permute parties. This proves the last assertion. $\square$

For a party set S , let

$$L_C(S) = \{\ell \in L_C : \text{supp}(\ell) \subseteq S\}.$$

The general stabilizer partial-trace formula specializes to

$$\rho_S^C = q^{-|S|} \sum_{\ell \in L_C(S)} W_\ell. \quad (2.3)$$

For later block calculations, write

$$C_X = C \oplus 0, \quad C_Z^\perp = 0 \oplus C^\perp, \quad L_C = C_X \oplus C_Z^\perp.$$

This formula is the bridge used in both the rigidity proof and the marginal invariants below. A Pauli-labelled quantity is not by itself an LU invariant; the invariant statements will always be expressed either as covariance of the reduced operators or as basis-free tensor contractions.

The GRS boundary has a simple geometric meaning. A dimension-three length-six generalized Reed–Solomon code is represented, up to column multipliers, by six points on a nonsingular conic in $\mathbb{P}^2$. We use “GRS state” for the state of any such code. The non-GRS pencil in Section 4 consists of arc matrices H_t away from that conic locus and from the other factors needed to keep the columns distinct and the displayed coordinates defined.

3 Full-Weyl marginals and LU rigidity

We first isolate the linear-algebra statement that recovers a diagonal tensor’s coordinate axes. It is the full-Weyl counterpart of the three-Pauli argument in [Rai99, VdNDDM05b].

Lemma 3.1 (Diagonal-tensor axes). Let $r \geq 3$, let $E_1, \dots, E_r$ be N -dimensional complex inner product spaces, and choose orthonormal bases $\{e_{ij} : 1 \leq j \leq N\}$. If

$$T = \sum_{j=1}^N \lambda_j e_{1j} \otimes \cdots \otimes e_{rj}, \quad \lambda_j \neq 0, \quad (3.1)$$

then the coordinate axes in every E_i are intrinsic to T . Consequently, a product of invertible linear maps carrying one tensor of the form (3.1) to another is monomial in the displayed bases.

Proof. For nonzero $x = \sum_j x_j e_{1j}^*$, contract the first factor against x . The result is

$$T_x = \sum_j \lambda_j x_j e_{2j} \otimes \cdots \otimes e_{rj}.$$

Flattening from E_2 to $E_3 \otimes \cdots \otimes E_r$ gives a matrix of rank $\#\{j : x_j \neq 0\}$. Thus T_x has rank one exactly when the ray of x is a dual coordinate axis. Product equivalence preserves this projective rank-one contraction locus. The inner products identify the recovered dual axes with the coordinate axes in the first factor. Apply the same argument in every factor. $\square$

Call an operator on $r \geq 3$ qudits *full-Weyl diagonal* if there is a q^2 -element set P , a distinguished element $0 \in P$, and bijections $f_i : P \rightarrow \mathbb{F}_q^2$ with $f_i(0) = 0$, for which

$$R = \sum_{v \in P} \lambda_v \bigotimes_{i=1}^r W_{f_i(v)}, \quad \lambda_v \neq 0. \quad (3.2)$$

The coefficients need not be equal. What matters is that the same q^2 -element index set runs bijectively through the complete local Weyl basis at every party. Neither P nor the bijections need be $\mathbb{F}_q$ -linear.

Proposition 3.2 (Full-Weyl marginal criterion). Let R and R' be full-Weyl diagonal operators on the same $r \geq 3$ parties, possibly with different indexing sets, bijections, and coefficients. If

$$R' = (U_1 \otimes \cdots \otimes U_r) R (U_1 \otimes \cdots \otimes U_r)^\dagger,$$

then every U_i is Clifford.

Proof. View R and R' as tensors in the local Hilbert–Schmidt spaces. After replacing W_v by the orthonormal basis $q^{-1/2}W_v$ and absorbing the common normalization into the coefficients, both have the form in Lemma 3.1, with $N = q^2$. Local conjugation acts invertibly on each factor, so it must permute the Weyl axes. It fixes the identity operator and hence the identity axis. It therefore permutes the nonidentity projective Weyl operators, which is exactly the single-qudit Clifford condition. $\square$

Corollary 3.3 (Full-Weyl marginal-cover rigidity). Let $|\psi\rangle, |\phi\rangle$ be n -qudit states and suppose a product unitary, after a party relabelling, carries $|\psi\rangle$ to $|\phi\rangle$ up to phase. If every party belongs to a set S of at least three parties for which both reduced operators ρ_S^ψ and ρ_S^ϕ are full-Weyl diagonal, then every local factor is Clifford.

Proof. Partial trace gives a product-unitary equivalence of the two reduced operators on each chosen S . Apply Proposition 3.2; the chosen sets cover all parties. $\square$

Proposition 3.4 (Stabilizer-AME support theorem). Let $|\psi\rangle$ be a stabilizer AME($2m, q$) state, and let A be any $m + 1$ parties. Then

$$|L(A)| = q^2.$$

Every nonidentity label in $L(A)$ has support exactly A , and for each $i \in A$ the coordinate projection

$$p_i : L(A) \longrightarrow \mathbb{F}_q^2$$

is bijective.

Proof. The projective label group L has order q^{2m} . Restrict its labels to the $m - 1$ parties outside A . The target has order $q^{2(m-1)}$, and the kernel is $L(A)$; hence

$$|L(A)| \geq q^{2m} / q^{2(m-1)} = q^2. \quad (3.3)$$

Fix $i \in A$. A label in the kernel of p_i is supported on the m -set $A \setminus \{i\}$. The AME condition and (2.2) exclude every such nonidentity stabilizer, so p_i is injective. Its codomain has order q^2 , giving the reverse inequality in (3.3) and hence bijectivity. The same no-small-support argument shows that every nonidentity element of $L(A)$ has full support A . $\square$

Proposition 3.5 (A stabilizer-AME marginal fixes the local Weyl axes). Let $|\psi\rangle, |\phi\rangle$ be stabilizer AME($2m, q$) states. If a product unitary on an $(m + 1)$ -set A carries ρ_A^ψ to ρ_A^ϕ , then every local factor on A is Clifford.

Proof. By (2.2),

$$\rho_A^\psi = q^{-|A|} \sum_{\ell \in L_\psi(A)} \widetilde{W}_\ell.$$

Index this sum by $P = L_\psi(A)$. Proposition 3.4 says that every coordinate projection $P \rightarrow \mathbb{F}_q^2$ is bijective. Relative to the fixed Weyl convention, each stabilizer lift contributes a nonzero coefficient. Thus ρ_A^ψ is full-Weyl diagonal in the sense of (3.2), and the same holds for ρ_A^ϕ . Apply Proposition 3.2. $\square$

Proof of Theorem 1.1. Absorb the party permutation by relabelling $|\phi\rangle$. Partial trace gives a product-unitary equivalence $\rho_A^\psi \sim \rho_A^\phi$ for every $(m+1)$ -set A . Given a party i , choose such a set containing it and apply Proposition 3.5. Thus every U_i is Clifford. $\square$

3.1 Minimum-support atlases and symmetry groups

The same support calculation recovers standard quantum-MDS structure. If $|A| \leq m$, the AME condition gives $L(A) = \{0\}$. If $|A| > m$, then $\rho_{A^c}^\psi$ is maximally mixed, so purity of the global state gives $\text{Tr}((\rho_A^\psi)^2) = q^{-(2m-|A|)}$. On the other hand, (2.2) and Weyl orthogonality give $\text{Tr}((\rho_A^\psi)^2) = q^{-|A|}|L(A)|$. Hence, for every party set A ,

$$|L(A)| = q^{2 \max(|A|-m, 0)}. \quad (3.4)$$

Möbius inversion is the stabilizer-state specialization of the quantum-MDS Shor–Laflamme distribution of Huber and Grassl [HG20, Theorems 8 and 9]; in particular the number of weight- $m+1$ projective stabilizers is $\binom{2m}{m+1}(q^2-1)$. We use one further consequence. If $|A| \geq m+2$ and $i \neq j$ lie in A , then the dimensions in (3.4), together with

$$L(A \setminus \{i\}) \cap L(A \setminus \{j\}) = L(A \setminus \{i, j\}),$$

give

$$L(A) = L(A \setminus \{i\}) + L(A \setminus \{j\}). \quad (3.5)$$

Descending induction shows that the $(m+1)$ -supported subgroups span the full label group. This is the additive quantum-MDS form of minimum-weight generation, not a separate weight-enumerator claim.

We now prove the atlas classification stated in the introduction. Regard each local label group

$$P_i = \mathbb{F}_q^2$$

as a $2e$ -dimensional trace-symplectic space over $\mathbb{F}_p$. For every $(m+1)$ -set A and $i, j \in A$, the support theorem defines the operator-pushing transition

$$M_A(i, j) = p_j p_i^{-1} : P_i \longrightarrow P_j.$$

For two states ψ, ϕ on the same labelled parties, a *symplectic atlas equivalence* is a tuple $F_i \in \text{Sp}(P_i)$ satisfying

$$F_j M_A^\psi(i, j) = M_A^\phi(i, j) F_i \quad \text{for every } A \text{ and } i, j \in A. \quad (3.6)$$

Here $\text{Sp}(P_i) = \text{Sp}_{2e}(\mathbb{F}_p)$ is the action induced by the standard additive local Clifford group.

For one state, define

$$\begin{aligned} \Gamma_\psi &= \{[U] \in \text{PU}(q^{2m}) : U = U_1 \otimes \cdots \otimes U_{2m}, U|\psi\rangle \in \mathbb{C}|\psi\rangle\}, \\ \mathcal{G}_\psi &= \{(F_i) \in \prod_i \text{Sp}(P_i) : F_j M_A(i, j) = M_A(i, j) F_i \text{ for all } A, i, j \in A\}. \end{aligned}$$

Theorem 1.1 and the Clifford sequence (2.0) give a well-defined homomorphism

$$\lambda_\psi : \Gamma_\psi \longrightarrow \mathcal{G}_\psi, \quad [U_1 \otimes \cdots \otimes U_{2m}] \longmapsto (\overline{U}_i)_i,$$

where $\overline{U}_i$ is the induced symplectic label map. A product decomposition is unique up to one-site scalars, which do not change $\overline{U}_i$. The injection $L_\psi \hookrightarrow \Gamma_\psi$ sends a label ℓ to the projective Pauli $[W_\ell]$.

Lemma 3.6 (Pauli phase correction). Let ψ and ϕ be stabilizer states with Lagrangian label groups L_ψ and L_ϕ . If a product Clifford U induces a symplectic label map carrying L_ψ onto L_ϕ , then there is a product Pauli W_v such that $W_v U|\psi\rangle \in \mathbb{C}|\phi\rangle$.

Proof. The state $U|\psi\rangle$ is stabilized on the same Pauli axes as $|\phi\rangle$, possibly with a different stabilizer character. The trace-symplectic pairing identifies the ambient label space modulo $L_\phi = L_\phi^\perp$ with the character dual of L_ϕ . Choose v whose pairing character is the inverse of the discrepancy. Conjugation by the product Pauli W_v cancels it. The corrected state and $|\phi\rangle$ therefore lie in the same one-dimensional joint stabilizer eigenspace. $\square$

Proof of Theorem 1.3. First fix the party labels. A block symplectic map $F = \bigoplus_i F_i$ carrying L_ψ to L_ϕ carries every supported subgroup $L_\psi(A)$ to $L_\phi(A)$. Writing these subgroups as graphs through any projection p_i gives exactly (3.6).

Conversely, suppose (3.6) holds. Fix A and $i \in A$. Every element of $L_\psi(A)$ is uniquely determined by its i -coordinate, and the equations with $j \in A$ show that its image under F is the unique element of $L_\phi(A)$ with i -coordinate $F_i p_i(\ell)$. Thus $FL_\psi(A) = L_\phi(A)$. Since the $(m+1)$ -supported subgroups generate the full label groups, $FL_\psi = L_\phi$.

Sequence (2.0) supplies local Clifford lifts of the F_i , and Lemma 3.6 supplies the product-Pauli correction. Hence an atlas equivalence lifts to an LC equivalence. The reverse implication follows from the induced label action, and Theorem 1.1 replaces LC by LU. A party permutation is absorbed by relabelling first.

For automorphisms, λ_ψ is surjective by the same lifting and phase-correction argument. Its kernel consists of projective product Paulis by (2.0). Such a Pauli fixes the state ray exactly when its label lies in $L_\psi^\perp = L_\psi$. Thus the displayed injection of L_ψ identifies it with $\ker \lambda_\psi$, proving exactness.

Finally suppose $q = p$. Form the directed multigraph with vertices the parties and an edge (A, i, j) carrying $M_A(i, j)$. Every pair of parties belongs to an $(m+1)$ -set, so the graph is connected. Choose a base b , a spanning tree, and tree transports $T_i : P_b \rightarrow P_i$, with $T_b = I$. For every edge define its based holonomy

$$H_A(i, j) = T_j^{-1} M_A(i, j) T_i \in \mathrm{GL}_2(q).$$

Compatibility propagates a base block F_b uniquely as $F_i = T_i F_b T_i^{-1}$; the edge (A, i, j) is then compatible exactly when F_b commutes with $H_A(i, j)$. Since $\mathrm{Sp}_2(\mathbb{F}_q) = \mathrm{SL}_2(q)$ is normal in $\mathrm{GL}_2(q)$, every propagated block remains symplectic. Evaluation at the base therefore identifies $\mathcal{G}_\psi$ with the holonomy centralizer inside $\mathrm{SL}_2(q)$. Different spanning trees generate the same based holonomy subgroup; changing the base conjugates that subgroup, and hence its centralizer, by a chosen transport. $\square$

3.2 Encoder and projective-symmetry consequences

Let $C \leq \mathbb{F}_q^{2m}$ be $[2m, m, m+1]_q$ MDS, and fix a party set S of size $m+1$. Shortening C on the $m-1$ coordinates outside S gives an $[m+1, 1, m+1]_q$ MDS code. The dual $C^\perp$ has

the same parameters, so its corresponding shortening does as well. Choose nonzero shortened words $c^S \in C$ and $h^S \in C^\perp$. Each has support exactly S , and

$$L_C(S) = \{(ac^S, bh^S) : a, b \in \mathbb{F}_q\}. \quad (3.7)$$

At a party $i \in S$, projection of this plane to the local Pauli labels is

$$(a, b) \mapsto (ac_i^S, bh_i^S),$$

which is invertible because both coordinates are nonzero. Thus all q^2 labels in $L_C(S)$ run bijectively through the full local Weyl basis at every party of S ; in particular, the $q^2 - 1$ nonidentity labels run bijectively through the nonidentity Weyl labels.

Proof of Corollary 1.2. The input marginal of each AME Choi state is $q^{-1}I$, so one-leg reshaping gives an isometry. The projection of its stabilizer label group onto the input Weyl-label space is surjective: otherwise a nonzero local label orthogonal to the image would lie in the maximal isotropic space $L^\perp = L$, contradicting the absence of a one-party stabilizer. The kernel therefore has q^{2m-2} elements and stabilizes the image, making it a one-logical-qudit stabilizer code. AME corrects erasure of any $m - 1$ outputs, so its distance is at least m ; the quantum Singleton bound gives equality. This is the standard perfect-tensor/quantum-MDS correspondence used in [Tan26, Theorems 3.5 and 3.6].

Let $|\Phi\rangle = q^{-1/2} \sum_x |x, x\rangle$, and use the Choi convention $|\psi\rangle = (I \otimes V_\psi)|\Phi\rangle$ and $|\phi\rangle = (I \otimes V_\phi)|\Phi\rangle$. Writing $U_{\text{phys}} = \bigotimes_i U_i$, the relation $U_{\text{phys}} V_\psi = V_\phi L$ gives

$$(I \otimes U_{\text{phys}})|\psi\rangle = (L^T \otimes I)|\phi\rangle.$$

Hence

$$((L^T)^{-1} \otimes U_{\text{phys}})|\psi\rangle = |\phi\rangle.$$

Theorem 1.1 makes every displayed local factor Clifford. Entrywise conjugation preserves the finite-field Weyl group, since $\overline{X(a)} = X(a)$ and $\overline{Z(b)} = Z(-b)$; it therefore preserves the Clifford normalizer. Taking adjoints preserves the normalizer as well, and $U^T = (\overline{U})^\dagger$. Thus transposition preserves the finite-field Clifford group, and L is Clifford. $\square$

Corollary 3.7 (Discrete product-unitary symmetry). Fix a stabilizer AME state $|\psi\rangle$ as in Theorem 1.1, and define the local-factor tuple groups

$$G = \{(U_i) \in U(q)^{2m} : (\bigotimes_i U_i)|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

$$\tilde{G} = \{((U_i), \pi) \in U(q)^{2m} \rtimes S_{2m} : (\bigotimes_i U_i)\pi|\psi\rangle \in \mathbb{C}|\psi\rangle\},$$

where S_{2m} permutes the local factors. Let $T = (S^1)^{2m}$ be the coordinatewise scalar subgroup and put $\Gamma = G/T$, $\tilde{\Gamma} = \tilde{G}/T$. Then

$$1 \longrightarrow T \longrightarrow G \longrightarrow \Gamma \longrightarrow 1, \quad 1 \longrightarrow T \longrightarrow \tilde{G} \longrightarrow \tilde{\Gamma} \longrightarrow 1$$

are short exact sequences with closed connected kernel and finite discrete quotient; both inclusions of T are continuous. In particular, G and $\tilde{G}$ are finite unions of connected components, each a translate of T . Thus the local-factor automorphism group modulo one-site scalar phases is finite and discrete, and its identity component before quotienting is exactly T . If Π is the subgroup of party permutations realized by $\tilde{G}$, there is a further exact sequence

$$1 \longrightarrow \Gamma \longrightarrow \tilde{\Gamma} \longrightarrow \Pi \longrightarrow 1.$$

This last extension splits exactly when the realized permutations admit a homomorphic choice of projective product-unitary lifts. It has a section-free outer action $\Pi \rightarrow \text{Out}(\Gamma)$: conjugation by any lift of $\pi \in \Pi$ gives an automorphism of Γ , and changing the lift changes that automorphism by an inner automorphism. For a normalized set-theoretic section $s(1) = 1$, write

$$s(\pi)s(\rho) = f_s(\pi, \rho)s(\pi\rho), \quad f_s(\pi, \rho) \in \Gamma.$$

Then $f_s(1, \pi) = f_s(\pi, 1) = 1$, and, with α_π denoting conjugation by $s(\pi)$,

$$f_s(\pi, \rho)f_s(\pi\rho, \nu) = \alpha_\pi(f_s(\rho, \nu))f_s(\pi, \rho\nu).$$

If another normalized section is $s'(\pi) = b(\pi)s(\pi)$, where $b(1) = 1$, then

$$f_{s'}(\pi, \rho) = b(\pi)\alpha_\pi(b(\rho))f_s(\pi, \rho)b(\pi\rho)^{-1}.$$

The order of these factors is essential because Γ need not be abelian. The extension splits exactly when this factor set can be changed to the constant identity factor set by such a normalized b .

Proof. Every local factor in a tuple from G or $\tilde{G}$ is Clifford. The projective single-qudit Clifford group over a finite field is finite, as is the party-permutation group. The kernel of projectivizing each local factor is exactly the torus of one-site scalar phases. That torus is the identity component, hence is closed; the finite Hausdorff quotient is discrete. Translation identifies every connected component with the scalar torus, so finitely many quotient representatives cover the group. Forgetting the local factors maps $\tilde{\Gamma}$ onto the realized permutation subgroup Π . Its kernel consists exactly of the fixed-party projective automorphisms, which gives the third sequence. A splitting is, by definition, a group-homomorphic right inverse; arbitrary or generator-by-generator lifts do not provide one. Conjugation by the middle group descends to the outer action because elements of the kernel act by inner automorphisms. Associativity of three chosen lifts gives the displayed factor-set identity. Substituting $s'(\pi) = b(\pi)s(\pi)$ gives the change-of-section formula without commuting any kernel factors. Finally, the factor set is identically one precisely when the section preserves multiplication, which proves the splitting characterization. $\square$

3.3 Exact MDS–CSS logical images and lifts

Proposition 3.8 (The diagonal-multiplier line). Let $E, F \leq \mathbb{F}_q^{2m}$ be linear $[2m, m, m+1]_q$ MDS codes, and put

$$\mathcal{D}(E, F) = \{s \in \mathbb{F}_q^{2m} : \text{diag}(s)E \subseteq F\}.$$

Every nonzero $s \in \mathcal{D}(E, F)$ has full coordinate support and $\text{diag}(s)E = F$. Consequently

$$\dim_{\mathbb{F}_q} \mathcal{D}(E, F) \leq 1, \quad \mathcal{D}(E, E) = \mathbb{F}_q(1, \dots, 1).$$

In particular, $\mathcal{D}(C, C^\perp)$ has dimension zero or one, and C is diagonally isodual exactly in the dimension-one case. When that line exists, any two nonzero witnesses differ by a unique nonzero scalar, so every ratio s_i/s_j is intrinsic.

Proof. Let $s \neq 0$, and write $D = \text{diag}(s)$. If D had support of size at most m , choose a word of E nonzero at a supported coordinate. Its image would be a nonzero word of F of weight at most m , contrary to the distance of F . Thus D has at least $m+1$ nonzero entries. Its kernel on E consists of words supported on at most $m-1$ coordinates, so the distance of E makes that kernel zero. Equal dimensions now give $DE = F$. A zero diagonal entry would force every word of F to vanish at that coordinate, which an MDS code does not, so s has full support.

If two multipliers were independent, a nonzero linear combination could be chosen to vanish at one coordinate, contradicting full support. Hence $\dim \mathcal{D}(E, F) \leq 1$. The all-ones vector belongs to $\mathcal{D}(E, E)$, proving the self-multiplier statement. Taking $F = C^\perp$ gives the remaining assertions. $\square$

Proof of Corollary 1.4. Write a fixed-party product-Clifford action on Pauli labels as

$$F_i = \begin{pmatrix} \alpha_i & \beta_i \\ \gamma_i & \delta_i \end{pmatrix},$$

and let $D_\alpha, D_\beta, D_\gamma, D_\delta$ be the diagonal matrices with the corresponding entries. Preservation of $L_C = C_X \oplus C_Z^\perp$ is equivalent to

$$\begin{aligned} D_\alpha C &\subseteq C, & D_\gamma C &\subseteq C^\perp, \\ D_\beta C^\perp &\subseteq C, & D_\delta C^\perp &\subseteq C^\perp. \end{aligned} \tag{3.8}$$

If the block at the chosen input is nondiagonal, then D_β or D_γ is nonzero. Proposition 3.8 and (3.8) give

$$D_\gamma C = C^\perp \quad \text{or} \quad D_\beta C^\perp = C.$$

The second equality can be inverted, so either one supplies a nonsingular diagonal S with $SC = C^\perp$. Consequently, on the non-isodual branch every input block is diagonal. Its determinant is one, and the common blocks $\text{diag}(a, a^{-1})$ always preserve L_C ; hence the exact linear factor there is T .

Now suppose

$$S = \text{diag}(s_1, \dots, s_{2m}), \quad SC = C^\perp. \tag{3.9}$$

The projective vector $[s_1 : \dots : s_{2m}]$, and hence every propagation ratio s_i/s_j , depends only on C .

For $\lambda, \mu \in \mathbb{F}_q$, put

$$F_i^-(\lambda) = \begin{pmatrix} 1 & 0 \\ \lambda s_i & 1 \end{pmatrix}, \quad F_i^+(\mu) = \begin{pmatrix} 1 & \mu s_i^{-1} \\ 0 & 1 \end{pmatrix}. \tag{3.10}$$

On the CSS label space $L_C = (C \oplus 0) + (0 \oplus C^\perp)$, the product of the lower blocks sends

$$(c, h) \mapsto (c, h + \lambda S c),$$

and the product of the upper blocks sends

$$(c, h) \mapsto (c + \mu S^{-1} h, h).$$

Thus both products preserve L_C and lift to product-Clifford tensor symmetries: Lemma 3.6 supplies the required product-Pauli corrections. Thus label-space preservation is sufficient here; no coherent choice of Clifford phases is being assumed. At any chosen input party j , their blocks run through all lower and upper elementary unipotents because $s_j \neq 0$. Together these generate $\text{SL}_2(q)$. Under the Choi correspondence, an input block A induces the logical block A^{-T} . This map is an automorphism of $\text{SL}_2(q)$, so every one-qudit logical symplectic action has a product-Clifford physical implementation. Logical Paulis have physical Pauli representatives, so the full projective one-qudit Clifford group $\mathbb{F}_q^2 \rtimes \text{SL}_2(q)$ is transversal.

Corollary 1.2 excludes every product-unitary logical action outside that group. Hence the displayed group is exact.

For a generalized or extended generalized Reed–Solomon code, the standard dual formula supplies (3.9), including the extended evaluation point at length $q+1$ [HP03, Sections 5.2–5.3].

Thus the GRS tower lies on the diagonally isodual branch, but the proof used no evaluation structure.

There are two different lift questions behind this projective statement. Fix an input coordinate j , put $t_i = s_i/s_j$, and write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(q).$$

Propagation to coordinate i is

$$\phi_i(A) = \begin{pmatrix} a & t_i^{-1}b \\ t_i c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & t_i \end{pmatrix} A \begin{pmatrix} 1 & 0 \\ 0 & t_i^{-1} \end{pmatrix}. \quad (3.11)$$

Thus every ϕ_i is a homomorphism, ϕ_j is the identity, and the elementary matrices in (3.10) satisfy all $\mathrm{SL}_2(q)$ relations coordinatewise. This calculation also applies at the extended evaluation point. It uses only the diagonal duality $SC = C^\perp$, not the evaluation formula for a GRS code.

For the present odd prime q , and more generally for any odd prime power, the finite Stone–von Neumann representation of the Heisenberg group H_q extends to a genuine representation of $\mathrm{SL}_2(q) \ltimes H_q$ [Gér76, Theorem 1]. Apply its Weil factor to each $\phi_i(A)$. The resulting tensor product is multiplicative in A , and exact Egorov covariance preserves the CSS stabilizer line. Consequently the one-qudit Clifford scalar extension, restricted to the linear $\mathrm{SL}_2(q)$ factor, splits coherently with the propagated tensor symmetries. The exceptional uniqueness clause at $q = 3$ in Gérardin’s theorem does not remove existence.

The affine extension does not split. Choose $a, b \in \mathbb{F}_q$ with $\chi(ab) \neq 1$, and put $X = X(a)$, $Z = Z(b)$. Their projective classes commute, whereas every choice of unitary representatives satisfies

$$ZX = \chi(ab)XZ, \quad \chi(ab) \neq 1.$$

Scalar changes of X and Z do not change their commutator. Hence the translation subgroup $\mathbb{F}_q^2$ has no homomorphic lift to the one-qudit unitary Clifford group. The coherent lift above is therefore $H_q \ltimes \mathrm{SL}_2(q)$, not a section of $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ through the scalar quotient. This is the Heisenberg central extension, not a residual metaplectic or Schur-multiplier obstruction on $\mathrm{SL}_2(q)$.

Neither conclusion decides the party-permutation extension in Corollary 3.7. Its quotient is the evaluation-set-dependent realized permutation group Π , and its kernel is the full fixed-party projective group Γ , rather than the scalar phases. Its splitting remains exactly the nonabelian factor-set triviality criterion stated there. $\square$

For a stabilizer state, the hypotheses of Corollary 3.3 hold whenever every party belongs to a support S of size at least three whose full supported stabilizer label subgroup is an $\mathbb{F}_q$ -linear plane and projects isomorphically onto the local Weyl-label plane at every party of S .

4 The non-GRS pencil and its quotient

Our first six-party application combines atlas classification with the rigidity theorem to turn a classical projective quotient into a complete LU coordinate over odd prime fields. Let H_t be the ordered matrix in (1.2), over a finite field of odd order, and put

$$G(t) = t^4 - 4t^3 + 7t^2 - 4t + 1.$$

The admitted non-GRS condition is

$$t(t-1)(t^2-t+1)(t^2-3t+1)G(t) \neq 0. \quad (4.1)$$

Factoring the twenty 3-by-3 column minors gives the first four factors, which keep the arc and quotient coordinates nondegenerate. The determinant for a conic through the six columns is a unit times G , so $G = 0$ is exactly the GRS boundary. Write $C_t = \ker H_t$ and $\Psi_t = \Psi_{C_t}$, and let A, B, z be as in (1.3).

Theorem 4.1 (Projective and Clifford quotient). For admitted t, u , the following are equivalent:

$$H_t \sim_{\text{proj}} H_u, \quad C_t \sim_{\text{mon}} C_u, \quad z(t) = z(u),$$

allowing arbitrary party permutations. If q is prime, these are also equivalent to $\Psi_t \sim_{\text{LC}} \Psi_u$.

Proof. Set

$$y(t) = \frac{(t-1)^2}{t}.$$

Then $A = -4t^2y$, $B = t^2(y^2 - 1)$, and

$$z = \frac{(y - y^{-1})^2}{16}.$$

It follows that $z(t) = z(u)$ exactly when

$$y(u) \in \{y(t), -y(t), y(t)^{-1}, -y(t)^{-1}\}. \quad (4.2)$$

Each of the four relations is realized by a direct projectivity and a party permutation. In order, one may use

relation	projectivity
$y(u) = y(t)$	$\text{diag}(1, (1-u)/(1-t), u/t)$
$y(u) = -y(t)$	$\text{diag}(1, (1-u)/(1-t), -u/t)$
$y(u) = y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (u-1)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}$
$y(u) = -y(t)^{-1}$	$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & (1-u)/t \\ 0 & -u/(1-t) & 0 \end{pmatrix}.$

The corresponding permutations are respectively the identity, (56), (35)(46), and (3546). All denominators and determinants are nonzero on the admitted locus.

Conversely, the ten signed products of complementary 3-by-3 brackets of H_t form the multiset

$$\{+A, +A, +A, -A, -A, -A, +B, +B, -B, -B\}. \quad (4.3)$$

Projectivity and independent column rescaling multiply all ten products by one common scalar; party permutation changes at most their common sign. On the admitted locus $A, B \neq 0$. If $A^2 \neq B^2$, the multiplicities three and two in (4.3) distinguish the two opposite pairs and recover $(B/A)^2 = z$. If $A^2 = B^2$, the multiset instead collapses to two opposite values, each of multiplicity five. That collapsed pattern is itself invariant under the common scaling and sign, and it is equivalent to $z = 1$. Thus the multiset recovers z in both cases, and projective equivalence forces equality of z . The equivalence with monomial codes follows from Section 2, and projective equivalence supplies an LC equivalence.

We now specialize the transition atlas from the rigidity argument to $m = 3$. Each four-party support identifies its two-dimensional stabilizer plane with the local Pauli-label plane at any party in that support. Overlapping supports therefore give coordinate changes between local Weyl frames, and their compositions around a loop give holonomy up to local symplectic conjugacy. By (3.5), these minimum-support planes generate the full stabilizer label group. The 450 quantities below are conjugacy data of this atlas, not an independent choice of scalar contractions.

For the remaining direction assume that q is prime. Then a local Clifford induces a block $F_i \in \mathrm{SL}_2(q)$ on each local Pauli-label plane. Put $K_T = L_{C_i}(T)$ for every four-set T . If $i, j \in T$, projection $p_i : K_T \rightarrow \mathbb{F}_q^2$ onto the Pauli-label plane at i is an isomorphism by (3.7). Define the transition

$$M_T(i, j) = p_j p_i^{-1} : (\mathbb{F}_q^2)_i \longrightarrow (\mathbb{F}_q^2)_j$$

and, for distinct four-sets T, U containing i, j , the holonomy

$$\mathrm{Hol}(T, U; i, j) = M_U(j, i) M_T(i, j). \quad (4.4)$$

Local symplectic blocks F_i send this endomorphism to $F_i \mathrm{Hol} F_i^{-1}$. Reversing T, U takes its inverse, and a party permutation only relabels the data. Thus the multiset of

$$r(M) = \mathrm{Tr}(M)^2 / \det M$$

over the $15 \cdot 6 \cdot 5 = 450$ choices is an LC and party-permutation invariant.

Substituting the shortened generators into (4.4), simplifying each of the 450 rational functions exactly, and collecting equal functions gives the following multiset. The paper-local certificate described in Section 7 replays both this symbolic enumeration and a separate direct-Lagrangian reconstruction:

value of $r(M)$	multiplicity
4	90
$-1/z$	144
$(2z+1)^2/z^2$	24
each root of $X^2 - 8X + (8 - 16z - 1/z)$	96.

(4.5)

Because $A, B \neq 0$ on the admitted locus, $z \neq 0$. The two roots are taken over an algebraic closure; their symmetric multiset descends to $\mathbb{F}_q$. The multiplicity 96 applies to each root before specialization; coincident values have their multiplicities added. The identity

$$4B(t)^2 + A(t)^2 = 4G(t)^2$$

shows that the excluded GRS locus is exactly $z = -1/4$. Away from it, the constant 4 bin is isolated and the two quadratic roots are distinct. The bin containing $-1/z$ can merge with neither, one, or both of the weight-24 and one weight-96 contributions, so its multiplicity is 144, 168, 240, or 264. No other bin has one of these multiplicities. Hence the multiset canonically recovers $-1/z$, including every modular collision, and LC equivalence forces equality of z . $\square$

The map $t \mapsto z$ has degree eight: the four transformations in (4.2), followed by the quadratic relation $t+t^{-1} = y+2$, account for its eight sheets. The signed refinement $w = B/A$, $z = w^2$, is not quantum data. Both even and odd explicit party permutations reverse w , and their projectivities lift to monomial local Cliffords.

Proof of Corollary 1.5. Theorem 4.1 gives all equivalences except LU. Every LC is an LU, while Theorem 1.1 turns every LU intertwiner between the states Ψ_t, Ψ_u into an LC intertwiner. $\square$

Proposition 4.2 (Frobenius-sector divisors). Let σ be an automorphism of a finite field of odd order and let t be admitted and non-GRS. Put

$$E_\sigma(t) = (\sigma(t) - t)(1 - \sigma(t)t), \quad D_\sigma(t) = ((1 - \sigma(t))(1 - t))^2 + \sigma(t)t.$$

Then

$$\frac{\sigma(t)}{(1 - \sigma(t))^2} = \frac{t}{(1 - t)^2} \iff E_\sigma(t) = 0. \quad (4.6)$$

If $h = (1 - \sigma(t))(1 - t)$ and $w = (-h, -h, 1, 1, -1, -1)$, then

$$H_{\sigma(t)} \text{diag}(w) H_t^\top = 0 \iff D_\sigma(t) = 0. \quad (4.7)$$

Moreover $D_{\text{id}}(t) = G(t)$, and

$$E_\sigma(t) = 0 \implies D_\sigma(t) \neq 0. \quad (4.8)$$

Finally G, A, B, z commute with σ , and σ preserves the admitted non-GRS locus.

Proof. Admission gives $t \neq 1$, and injectivity of σ gives $\sigma(t) \neq 1$. Cross multiplication in (4.6) gives

$$s(1 - t)^2 - t(1 - s)^2 = (s - t)(1 - st)$$

with $s = \sigma(t)$. Direct multiplication in (4.7) makes eight entries vanish identically and leaves $-2D_\sigma(t)$ in the last entry; odd characteristic gives the equivalence. Setting σ to the identity yields $D_{\text{id}} = G$.

If $E_\sigma(t) = 0$, then either $\sigma(t) = t$ or $\sigma(t)t = 1$. In the first case $D_\sigma(t) = G(t)$. In the second, multiplying $D_\sigma(t)$ by t^2 gives $G(t)$. Admission excludes $t = 0$ and $G(t) = 0$, proving (4.8). The equivariance statements follow by applying σ to the defining polynomial and rational expressions. $\square$

The prime-field hypothesis is necessary for this scalar classification. Over $\mathbb{F}_{p^e}$, the basis permutation $|x\rangle \mapsto |x^p\rangle$ is a local Clifford and sends Ψ_t to Ψ_{t^p} , while $z(t^p) = z(t)^p$ need not equal $z(t)$. Concretely, in $\mathbb{F}_{25} = \mathbb{F}_5[s]/(s^2 - 2)$, both s and $-s = s^5$ are admitted, but

$$z(s) = 3 + 3s, \quad z(-s) = 3 + 2s.$$

For each field automorphism σ , Proposition 4.2 gives the collision and diagonal-isodual divisors relating t to $\sigma(t)$. It does not show that an arbitrary additive local Clifford determines a nontrivial field automorphism, nor does it construct a local Clifford from a Galois match of z . Those two representation-theoretic bridges are not claimed here, so neither is a full extension-field LC/LU classification.

5 Diagonal isoduality and the logical-Clifford phase

The second application uses the atlas gauges to determine the exact fixed-party transversal logical group of an MDS-CSS encoder. Throughout this section q is an odd prime. Regard any one party of an $\text{AME}(6, q)$ stabilizer tensor as an input and the other five as outputs. This is a $[[5, 1, 3]]_q$ encoding isometry. In prime local dimension, the symplectic action of a one-qudit Clifford is a block in $\text{SL}_2(q)$, so the kernel of the party action on Clifford tensor symmetries is

$$\mathcal{K}_C = \{(F_1, \dots, F_6) \in \text{SL}_2(q)^6 : (\bigoplus_i F_i) L_C = L_C\}.$$

For $q = p^e$ with $e > 1$, the full local Clifford group acts instead through $\text{Sp}_{2e}(\mathbb{F}_p)$; the displayed field-linear block group is then not the full local Clifford kernel. For an encoder view, the *fixed-party logical symplectic image* means the image of $\mathcal{K}_C$ on its input block. Corollary 1.4 identifies the all-length phase boundary as diagonal isoduality. For six-arcs, classical self-association turns that intrinsic code condition into the conic criterion below.

Lemma 5.1 (Fixed-label self-association). Let H be a rank-three 3×6 matrix whose columns are a six-arc, and put $C = \ker H$. There is a nonsingular diagonal matrix S with $SC = C^\perp$ if and only if the six columns of H lie on a conic.

Proof. Let G be a 3×6 generator matrix for C . Its columns are the labeled Gale transform of the columns of H . Since $C^\perp = \text{row}(H)$,

$$SC = C^\perp \iff \text{row}(GS) = \text{row}(H) \iff AGS = H \text{ for some } A \in \text{GL}_3(q).$$

Thus the condition is precisely fixed-label projective equivalence to the Gale transform. This fixed locus can be identified directly over the finite field. Put

$$V_i = (x_i^2, y_i^2, z_i^2, x_i y_i, x_i z_i, y_i z_i)$$

and let V be the 6-by-6 matrix with rows V_i . Its determinant vanishes exactly when a nonzero quadratic form contains all six columns. Every five rows of V are independent: a relation supported on at most five points would give $H_I D H_I^T = 0$ for a nonsingular diagonal D on its support I ; for $|I| \geq 3$, this would put the three-dimensional row space of $H_I D$ inside $\ker H_I$, which has dimension at most two, while smaller supports are immediate. Hence any relation $\sum_i d_i V_i = 0$ has every $d_i \neq 0$.

Writing $D = \text{diag}(d_i)$, that relation is $HDH^T = 0$. Therefore $\text{row}(HD) \subseteq \ker H = C$, and equality follows from dimension three. Thus $CD^{-1} = C^\perp$. Conversely, a nonsingular diagonal equivalence $SC = C^\perp$ gives $HS^{-1}H^T = 0$, hence a relation among the rows of V and a conic through the six columns. The conic is irreducible because the columns form an arc. This is the finite-field fixed-label form of the classical six-point self-association theorem [HMSV08, Section 2.3]. $\square$

Theorem 5.2 (Fixed-party logical phase). For an equal-phase CSS state from a six-arc over an odd prime field, the fixed-party logical symplectic image is

$$\begin{cases} \text{SL}_2(q), & \text{if the arc is conic/GRS,} \\ T = \{\text{diag}(a, a^{-1}) : a \in \mathbb{F}_q^\times\}, & \text{if the arc is nonconic.} \end{cases} \quad (5.1)$$

Thus every encoder view of a GRS tensor realizes the full one-qudit logical symplectic group from fixed-party symmetries. Off the GRS locus, the fixed-party logical symplectic image is the split torus. At a chosen anchor leg, any party-moving isoduality that exchanges the CSS X - and Z -spaces transports the encoder view by the nontrivial Weyl-group element, so the possible anchor blocks generate the normalizer of T together with T . This is an action on one fixed encoder only when the isoduality fixes its input leg.

Proof. Corollary 1.4 gives $\text{SL}_2(q)$ exactly on the diagonally isodual branch and T otherwise. Lemma 5.1 identifies diagonal isoduality with the conic locus. At a chosen anchor leg, a party-moving isoduality exchanges the CSS X - and Z -axes and hence represents the nontrivial Weyl-group element normalizing T . It acts on one fixed encoder only when the party motion fixes that encoder's input. $\square$

Adding logical Paulis recovers the exact fixed-party projective logical groups in Corollary 1.4.

This is an operational distinction, not a classical automorphism count. The fixed-party non-GRS group has order $q^2(q-1)$, while the GRS group has order $q^2|\text{SL}_2(q)|$.

The MDS property also makes minimum operator pushing uniform. For any input party and nonidentity Pauli label, each of the ten four-sets containing that party gives a unique stabilizer representative supported there. Removing the input leaves a weight-three output Pauli. Thus each input Pauli has exactly ten minimum-support pushes, although the symmetry orbits of those pushes can distinguish finer projective classes. This uses only the projection isomorphism for the shortened planes, not the finite group calculation.

6 Scalar invariants and generic blindness

The operator-valued marginal mechanism retains local Weyl axes. Fixed-copy scalar contractions retain less: on algebraic families they are generically constant and can distinguish parameters only on rank-jump strata. We make that limitation precise here; Appendix A records two explicit separators on exceptional strata.

For r ket copies and r bra copies, choose a permutation $\sigma_i \in S_r$ at each party and contract ket copy a with bra copy $\sigma_i(a)$. If G generates a d -dimensional code, let $M_\sigma(G)$ be the homogeneous matching system

$$(u_a G)_i = (v_{\sigma_i(a)} G)_i \quad (1 \leq i \leq n, 1 \leq a \leq r).$$

Counting its solutions gives the normalized contraction

$$I_\sigma(\Psi_C) = q^{dr - \text{rank } M_\sigma(G)}. \quad (6.1)$$

Proposition 6.1 (Generic constancy). Fix r and a field k of cardinality q , which is also the local Hilbert-space dimension under the code–state dictionary. Fix an irreducible reduced parameter variety X over k , and a regular generator-matrix chart $G(x)$ of constant code dimension. Every degree- (r, r) permutation contraction of the associated equal-phase states has one value at every k -point of some dense open subscheme $U \subseteq X$ defined over k . This asserts no existence of k -points in U : if $U(k) = \emptyset$, the finite-field pointwise conclusion is vacuous. The same holds for party-orbit sums and linear combinations. After evaluating the states as vectors in $(\mathbb{C}^q)^{\otimes n}$, in the stable range $q \geq r$, every scalar LU-invariant of bidegree (r, r) is generically constant in this sense. For a reducible family, the statement holds separately on each irreducible component.

Proof. Formula (6.1) depends only on the rank of a matrix whose entries are regular functions over k . A maximal nonzero minor shows that this rank is constant on a dense open subscheme defined over k . There are only finitely many diagrams at fixed r , so their generic-rank opens have a common dense intersection because X is irreducible. This proves the geometric generic-rank assertion; the explicit caveat in the statement is necessary when passing to finite-field rational points. Over the complex vector space $(\mathbb{C}^q)^{\otimes n}$, applied pointwise to each code state, the first fundamental theorem for unitary invariants says that local copy-permutation diagrams span the bidegree- (r, r) invariants; when the local Hilbert-space dimension q is at least r , there are no additional dimension relations among these diagrams [CŠ06, Section 2]. $\square$

Thus no fixed copy number can recover a nonconstant pencil coordinate on the generic locus. A scalar witness can still detect a special rank-jump stratum, as the four-copy example in Appendix A does. In contrast, the four-party operator tensor (3.2) retains the Weyl axes and forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

7 Verification and scope

The rigidity theorem, the arc–AME dictionary, and generic constancy are conceptual proofs in the text. The projective quotient uses displayed bracket identities and projectivities. Its transition-map invariance and collision argument are proved in Section 4, while the 450 exact rational functions are certificate-supported. Section 3 proves the general MDS diagonal-multiplier criterion. Section 5 combines it with Gale duality to obtain the six-arc phase theorem. The party groups, factor sets, complements, and torus actions in Corollary C.1 are certificate-supported finite examples, not inputs to the uniform theorem. Appendix A proves

the marginal trace/rank and incidence bridges; the H_3 determinants and $q = 13$ contractions are certificate-supported. Appendix B proves the systematic transport reduction; the cycle-cover determinants, assignment exhaustion, exact minors, and double-coset support sets are certificate-supported.

The reproducibility package is in `supplement/`. From that directory,

```
python3 verify.py          checks all seventeen byte counts and SHA-256 digests,
python3 verify.py --replay  regenerates all eight canonical JSON certificates in memory.
```

The full replay uses only the Python standard library, deterministic integer, rational, polynomial, and finite-field arithmetic, and canonical JSON serialization. It includes separately implemented checks appropriate to each claim: CSS versus direct-Lagrangian shortening, symbolic versus finite projectivity, chord concurrency versus stabilizer ranks, quotient minors versus finite-field elimination, and cycle-cover versus fraction-free transport determinants. These checks belong to one paper-owned Python package. They provide independent implementations within that package, not an independent external reproduction. The exact files, entry points, hashes, domains, expected outputs, and negative scopes are recorded in `supplement/EVIDENCE.md` and `supplement/EVIDENCE-MANIFEST.json`. On the reference machine the digest check takes under a second, the full evidence replay about six minutes, and each Lean aggregate under ten seconds.

From the repository root, the exact formal replay is

```
python3 lean/scripts/lean-build-queue.py run \
  RelativeConicArcs.Gates.AMELUAggregate \
  RelativeConicArcs.Gates.AMELUAggregateAxioms \
  --profile single --threads 1 \
  --aggregate RelativeConicArcs.Gates.AMELUAggregateAxioms
```

The checked release is then reproduced from `papers/ame_lu/` by

```
make release-check
```

The release identifier is content based: `release/RELEASE-MANIFEST.json` records the SHA-256 digest of every public byte and every file in the recursive formal dependency closure, including the Lean toolchain and build definitions. The manifest, rather than a mutable working-tree name, is the immutable identity of the companion release.

Claim-to-trust map. LU-to-LC rigidity, atlas classification, and the transversal no-go. The manuscript gives complete proofs. Lean kernel-checks the support, transition, marginal, covariance, axis-recovery, and holonomy cores; the manuscript proves the exact projective atlas extension.

Pencil and six-arc classification. The manuscript proof uses exact certificates for the finite holonomy calculation. The Lean interface is conditional and names its geometric and holonomy inputs.

H_3 and $q = 13$ separators. The manuscript proof uses exact certificates for the determinants and contraction ranks. The finite graph core is unconditional in Lean; the trace/covariance bridges and exact ranks are named inputs.

Diagonal-isodual logical groups. The manuscript gives the proof, and the multiplier theorem is unconditional in Lean. The formal exact-carrier terminal names the propagation interface as a hypothesis.

Twelve computed party-extension splittings and transport. Exact certificates supply the finite complements and transport matrices. Lean proves the abstract splitting algebra once a complement is supplied.

The corresponding certificate entry points and formal theorem names appear in the claim crosswalk in `supplement/EVIDENCE.md`.

The trusted boundary is Python integer arithmetic, the exact-arithmetic implementations in the generators, deterministic Gaussian elimination, and the mathematical dictionaries connecting arcs, codes, stabilizers, and tensor contractions. The certificates check the bounded enumerations and symbolic identities that they record. They do not prove the conceptual correspondence between the implemented objects and the paper definitions; that correspondence is supplied in the preceding sections.

The Lean 4 artifact uses toolchain `v4.32.0-rc1`; no aggregate module uses `sorry` or an unsafe declaration. Its aggregate contains unconditional formal proofs of the support squeeze, minimum-support generation, marginal and axis-recovery cores, atlas equivalence, abstract holonomy centralizers, the Choi/transversal bridge, the diagonal-multiplier theorem, and the abstract projective-extension algebra. The ordinary kernel-checked declarations use only `propext`, `Classical.choice`, and `Quot.sound`; three finite graph cardinalities use `native_decide` and are reported separately by the axiom terminal. All other finite evidence is replayed by the Python supplement rather than imported as a Lean axiom.

The geometric classification, exact logical propagation, finite contractions, concrete party complements, and transport-orbit inputs enter conditional interfaces. The manuscript proves or the supplement certificate-checks those inputs as recorded in the claim-to-trust map above; importing the aggregate does not turn them into unconditional Lean theorems. The exact module-to-claim crosswalk and each named hypothesis are recorded in `supplement/EVIDENCE.md`.

The extension-field scalar statements in Proposition 4.2 are unconditional in Lean. The aggregate audit checks the frame-ratio factorization, explicit Gale multiplier and odd-characteristic zero criterion, same-exponent sector disjointness, and automorphism equivariance of G, A, B, z and the admitted locus. Its separate Galois- z orbit terminal is conditional on two named bridges: extraction of a projective Frobenius sector from an additive Clifford equivalence, and construction of a Clifford from a Galois match. Neither bridge is used as a paper theorem.

Finally, none of the results classifies nonlinear orthogonal arrays, arbitrary nonstabilizer AME tensors, or stabilizer states that are not AME. We do not claim that four copies are globally minimal, that $z = 2, 4/9$ are orbit exceptions, or that a fixed copy degree separates all classes as q varies. The logical-group theorem and the quantum part of the pencil classification are restricted to odd prime fields. Over extension fields, Frobenius basis permutations are additional local Cliffords. The scalar z is Frobenius-covariant, and Proposition 4.2 determines the two divisors attached to each field automorphism, but the two additive-Clifford bridges needed for a full orbit classification are outside the theorem package. The minimum-support atlas theorem remains exact over extension fields, with local blocks in $\mathrm{Sp}_{2e}(\mathbb{F}_p)$; its holonomy form requires both a centralizing base block and the propagated symplectic membership conditions, and does not imply a semilinear classification. Theorem A.2 is restricted to the stated odd residue-field reductions, and the detector exceptions described in Appendix B do not alter the characteristic-uniform rigidity theorem.

8 Conclusion

An $(m + 1)$ -party marginal of a stabilizer $\mathrm{AME}(2m, q)$ state retains the full configuration of local Weyl axes. Rank-one contractions recover those axes intrinsically, so every product-unitary intertwiner is Clifford factor by factor. The minimum-support atlas is the structural completion of this rigidity mechanism: it records how the recovered frames fit together and classifies the residual Clifford equivalence exactly.

The applications expose two consequences of that structure. For the equal-phase linear MDS–CSS states over odd prime fields considered here, diagonal isoduality is exactly the boundary between the full projective logical $\mathbb{F}_q^2 \rtimes \mathrm{SL}_2(q)$ group and the affine split-torus subgroup. On the six-party pencil over odd prime fields, the degree-eight projective quotient z is also the complete LU coordinate. Fixed-copy scalar contractions, by contrast, discard the recovered axes and are generically constant on irreducible code families.

Over extension fields, the atlas classification remains exact with local blocks in $\mathrm{Sp}_{2e}(\mathbb{F}_p)$, but the two representation-theoretic bridges from additive Clifford maps to Frobenius sectors and back are not proved here. Determining when an extension-field atlas admits a semilinear canonical form is the remaining exact classification problem.

A Explicit scalar certificates

This appendix records two explicit scalar LU separators. They are not used in the uniform rigidity, atlas, or logical-group results; they show what finite-copy scalar contractions can still detect on exceptional strata despite the generic blindness proved in Section 6.

A.1 A marginal-moment separator

For a four-party set T , let ρ_T be the reduced state and embed it into all six parties as

$$A_T = \rho_T \otimes I_{T^c}.$$

For an unordered triple $\{T, U, V\}$ of distinct four-sets, the number $\mathrm{Tr}(A_T A_U A_V)$ is invariant under arbitrary product unitaries; party permutation merely permutes the 455 triples. By (2.3),

$$\mathrm{Tr}(A_T A_U A_V) = q^{-\mathrm{rank}(L_C(T) + L_C(U) + L_C(V))}. \quad (\text{A.1})$$

Here and below, ranks of label spaces and matching systems are over $\mathbb{F}_q$. Indeed, the three stabilizer sums have total normalization q^{-12} . Only triples of labels with sum zero survive the full trace; isotropy of L_C removes multiplication phases, and the kernel of the addition map from the three two-planes has size q^{6-r} , where r is the $\mathbb{F}_q$ -rank in (A.1).

Index a four-set by its omitted pair, an edge of K_6 . The rank in (A.1) is four exactly when the three corresponding chord lines concur in both the six-arc $\mathcal{A}$ and its Gale dual $\mathcal{A}^*$, represented by any kernel basis of the arc matrix. To see this, split each shortened CSS plane into its one-dimensional $X(C)$ and $Z(C^\perp)$ parts. For three omitted edges, the three shortened X -lines span dimension two precisely when the corresponding chords of $\mathcal{A}^*$ concur; the Z -lines give the same criterion for $\mathcal{A}$. Each span has dimension at least two, so their sum has rank four exactly when both concurrences hold.

Sixty triples are stars and concur universally. Any other three-edge graph that contains two adjacent edges but is not a star has two chords meeting at their common arc point and a third chord missing it, since the columns form an arc. The only remaining graph type is a perfect matching. Hence

$$\#\{q^{-4} \text{ moments}\} = 60 + b(\mathcal{A}, \mathcal{A}^*). \quad (\text{A.2})$$

where b counts perfect matchings concurrent in both configurations.

Lemma A.1 (Concurrent matchings on a conic). Let S be six distinct points on a nonsingular conic over a finite field k of odd characteristic other than 5. At most six perfect matchings of S have concurrent joining chords.

Proof. Identify the conic with $\mathbb{P}^1$. Projection from a point off the conic pairs the points in every fiber and hence induces a projective involution; conversely, the chords joining the three two-element orbits of a projective involution meet at its center. Thus the concurrent matchings are exactly the fixed-point-free involutions in

$$H = \text{Stab}_{\text{PGL}_2(k)}(S).$$

The correspondence is injective because an element of $\text{PGL}_2(k)$ is determined by its action on three points. Moreover, H acts freely on ordered triples of distinct points of S , so $|H| \mid 6 \cdot 5 \cdot 4 = 120$.

Dickson's classification, including the p -regular cases [Dic01, §§239–261], and Faber's p -irregular classification [Fab23, Theorem 1.1] now leave the following possibilities. The last column bounds involutions having no fixed point on this six-set:

H	fixed-point-free involutions
cyclic	≤ 1
dihedral	≤ 6
A_4	≤ 3
S_4	≤ 6
A_5 in its faithful degree-six action	0
C_3 or S_3 (char $k = 3$)	≤ 3 .

For the A_5 row, each involution fixes two of the six points. In the dihedral row, if the rotation order exceeded six, every orbit away from the two rotation-fixed points would have size at least that order, so no invariant six-set could exist; for rotation order at most six the two reflection classes together contribute at most six fixed-point-free elements (and the possible central half-turn contributes only when one reflection class has fixed points). In the characteristic-three semi-elementary case, the normal 3-group has order 3, while its cyclic complement embeds in $\text{Aut}(C_3)$, giving only C_3 or S_3 . The subfield groups whose order divides 120 are $\text{PSL}_2(3) \cong A_4$ and $\text{PGL}_2(3) \cong S_4$, already listed. The remaining characteristic-five subfield possibilities are excluded by the hypothesis. Every row is therefore bounded by six. $\square$

Theorem A.2 (Uniform H_3 /GRS separation). Let $\tau^2 = \tau + 1$, and let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[\tau]$ of odd residue characteristic for which the reduction of the integral H_3 (icosahedral) sextuple

$$((0, 1, 1 - \tau), (0, 1, \tau - 1), (1, 1 - \tau, 0), (1, \tau - 1, 0), (1, 0, -\tau), (1, 0, \tau))$$

is a six-arc. Write $k = \mathbb{Z}[\tau]/\mathfrak{p}$. If the reduced arc is non-GRS, put $q = |k|$. Its equal-phase state is not LU-equivalent, even after party permutation, to any six-point GRS state over k . (The hypothesis is vacuous in characteristic five, where the reduction is itself GRS.)

Proof. Exact integral chord determinants give ten common concurrent matchings for the H_3 arc and its Gale dual. The other five determinant norms are -64 and -4 , so the count is exactly ten in every residue reduction covered by the theorem. This is an exact fifteen-matching calculation in $\mathbb{Z}[\tau]$, not a field sample. The H_3 moment multiset therefore contains 70 copies of q^{-4} .

Lemma A.1 gives at most six concurrent matchings on a conic. The bound is attained whenever $\text{PGL}_2(q)$ contains an octahedral S_4 , by the corresponding six-set: the six half-turns about axes through pairs of opposite edges act without fixed vertices. Thus a GRS state has at most 66 copies of q^{-4} , and the margin $70 > 66$ is sharp for this argument. In characteristic five no subgroup bound is needed: $X^2 - X - 1 = (X - 3)^2$, so the residue of τ is 3, and direct substitution gives $G(\tau) = 0$. The H_3 reduction is therefore GRS, contrary to the theorem's non-GRS hypothesis. Equation (A.1) is LU-invariant, so the gap proves the claim. $\square$

The five nonconcurrent H_3 matchings form a one-factorization of K_6 . The marginal incidence recovers its S_5 stabilizer but forgets the even A_5 orientation. This is necessary: full local Fourier transform sends Ψ_C to $\Psi_{C^\perp}$, and an integral isoduality returns the H_3 code through an odd pentad permutation in every odd reduction. The orientation is an LC-forgettable CSS electric/magnetic marking, not a separate LU class.

A.2 The exact dimension-thirteen four-copy witness

Use the permutation contraction I_σ and matching system $M_\sigma(G)$ from (6.1).

Theorem A.3 (Exact four-copy separator at $q = 13$). The two admitted $q = 13$ pencil classes with $z = 4$ and $z = 12$ are not local-unitary equivalent, even after arbitrary party permutation.

Proof. At $q = 13$, the admitted pencil has two classes in the same complete three-copy and marginal-moment bucket, with $z = 4$ and $z = 12$. Use the explicit four-copy pattern

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

For $\pi \in S_6$, put $(\pi\sigma)_i = \sigma_{\pi^{-1}(i)}$, and sum (6.1) over all party permutations:

$$J_\sigma(\Psi) = \sum_{\pi \in S_6} I_{\pi\sigma}(\Psi).$$

Over $\mathbb{F}_{13}$, the $z = 4$ class has rank 21 for all 720 terms. The $z = 12$ class has rank 20 for 192 terms and rank 21 for 528. Therefore

$$J_\sigma(z = 4) = 720 \cdot 13^{-9}, \quad J_\sigma(z = 12) = 3024 \cdot 13^{-9}. \quad (\text{A.3})$$

This proves arbitrary-LU inequivalence directly. The complete two- and three-copy contraction sets do not separate this pair; (A.3) makes no claim that four copies are globally minimal. $\square$

Over $\mathbb{Q}(t)$, the matching matrices of every fixed four-copy diagram have generically constant rank. Consequently, after finite-field specialization on the common generic-rank open, their contraction values are constant and do not recover z generically. The witness (A.3) detects a rank-jump stratum. In contrast, the four-party operator tensor (3.2) retains the Weyl axes and forces Cliffordness. This is the precise distinction between scalar blindness and marginal covariant rigidity.

B Transport calculation for the four-copy witness

We now explain the rank jump behind (A.3). Fix the six copy permutations

$$\sigma = (1, (03)(12), (23), (02)(13), (01)(23), (021)).$$

Let V be the three-dimensional message space and let $g_i : V \rightarrow k$ be the six coordinate covectors. Form a bipartite four-sheeted cover with vertices L_a, R_b , $0 \leq a, b < 4$, and a color- i edge from L_a to $R_{\sigma_i(a)}$. Put V at each vertex and k at each edge, using $g_{p(i)}$ as both restriction maps for a party assignment $p \in S_6$. A global section is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{p(i)}(x_a - y_{\sigma_i(a)}) = 0. \quad (\text{B.1})$$

on all 24 edges. The three-dimensional constant-section space is always present. Gauging it away leaves the 24-by-21 matching matrix in (6.1).

Let $U = k^4/k(1, 1, 1, 1)$, and let ρ_i be the action of σ_i on U . For each p , the MDS property makes $g_{p(3)}, g_{p(4)}, g_{p(5)}$ a basis of V^* . Define Q_p by

$$g_{p(i)} = \sum_{k=0}^2 (Q_p)_{ik} g_{p(3+k)} \quad (0 \leq i < 3). \quad (\text{B.2})$$

Thus Q_p is the left block obtained by systematizing the party-reordered generator matrix on its last three columns; no pivot can vanish because every three coordinate covectors are independent. Put $\tilde{Q}_p = 2(t-1)Q_p$. This scalar is a unit on the admitted locus. For the identity assignment, the polynomial representative is

$$\tilde{Q}_1 = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix}, \quad \begin{aligned} a &= -2(t-1)^2, & b &= -(t^2 - t + 1), \\ c &= -(t^2 - 3t + 1), & d &= -2(t-1). \end{aligned}$$

If $Y_k \in U$ are the three free copy vectors, then (B.1) reduces to the 9-by-9 block transport operator

$$\tilde{T}_p = ((\tilde{Q}_p)_{ik}(\rho_i - \rho_{3+k}))_{i,k=0}^2. \quad (\text{B.3})$$

The dimension identity

$$21 - \text{rank } M_{\pi\sigma}(G(t)) = 9 - \text{rank } \tilde{T}_p \quad (\text{B.4})$$

holds over the field k , where $p = \pi$ in the convention of Section 6. It follows by quotienting the three-dimensional constant-section kernel and then eliminating the three free systematic coordinates. Multiplying the resulting operator by the unit $2(t-1)$ does not change its rank. All steps are invertible on the admitted six-arc locus.

Theorem B.1 (Transport divisor). For a parameter on the admitted odd non-GRS pencil, there exists a party assignment $p \in S_6$ for which the sheaf (B.1) has a nonconstant section if and only if

$$(B^2 - 2A^2)(9B^2 - 4A^2) = 0,$$

or equivalently

$$(z-2)(9z-4) = 0. \quad (\text{B.5})$$

At a generic point above $z = 2$, exactly 96 party assignments have rank 20; at a generic point above $z = 4/9$, exactly 192 do. Every other assignment has rank 21.

Proof. For the identity systematic chart, the coefficients satisfy

$$a = b + c, \quad d^2 = -2a, \quad A = a(c - b), \quad B = bc.$$

Common copy relabelling, a change of basis in V , and rescaling a coordinate covector act on $\tilde{T}_p$ by invertible row and column operations. Modulo these operations, the 720 party assignments split into boundary types and three nonboundary relative-cover types. Here the type records the six relative copy permutations $\rho_j^{-1}\rho_i$, equivalently the even-cycle pattern in each pair of colored matchings. Expanding a representative determinant by signed cycle covers gives

$$\begin{aligned} P_- &= AB(3B - 2A), \\ P_+ &= AB(3B + 2A), \\ P_2 &= A(B^2 - 2A^2). \end{aligned}$$

With integral bases the determinants are respectively

$$64d^3 AB(3B - 2A), \quad 64d^3 AB(3B + 2A), \quad 64d^3 A(B^2 - 2A^2).$$

All omitted factors are units on the admitted odd pencil. Every assignment outside these three types has a determinant that is a unit there. The signed cycle-cover ledger, the exhaustion of the 720 assignments, and a fraction-free determinant path are recorded in the paper-local certificate. The two signed types lie above $w = B/A = \pm 2/3$, hence $z = 4/9$; the axial type lies above $z = 2$. Fraction-free elimination also gives a nonzero 8-minor on each component, so the generic kernel is exactly one-dimensional.

The row/column equivalences organize the exceptional assignments into double cosets. The axial stabilizers have orders 48 and 16, with intersection of order 8, so its support has size

$$48 \cdot 16/8 = 96.$$

For either signed type the corresponding orders are 24, 48, and 6, giving

$$24 \cdot 48/6 = 192.$$

The two signed types are distinguished by the sign of w , while the axial cycle pattern differs from both; hence their support sets are disjoint. Explicit representatives, stabilizer intersections, and complete support sets are in the certificate. Equation (B.4) turns the one-dimensional transport kernels into rank 20 of the matching matrix. $\square$

The exceptional arithmetic belongs to this detector, not to Theorem 1.1. The identity $4B^2 + A^2 = 4G^2$ gives, respectively,

char k	specialized identities
3	$B^2 - 2A^2 = G^2, \quad 9B^2 - 4A^2 = -A^2,$
5	$9B^2 - 4A^2 = 4G^2,$
7	$9B^2 - 4A^2 = 2(B^2 - 2A^2).$

Thus in characteristic 3 both admitted components disappear; in characteristic 5 the $z = 4/9$ component lies on the excluded GRS boundary; and in characteristic 7 the two components merge. The disjoint 96- and 192-element supports then give the exact rank histogram: 288 assignments have rank 20, and 432 have rank 21. Characteristics 11, 13, 41 introduce only pullback ramification, with no further rank drop. These reductions and the nonzero 8-minors that control them are replayed in the certificate.

The divisor records where one scalar contraction changes rank. It is not an LU-orbit boundary and, by Proposition 6.1, cannot be promoted to a generic coordinate.

C Computed party-permutation extensions

This optional appendix records finite examples of the projective extension in Corollary 3.7. It may be skipped without affecting the uniform rigidity, atlas, or transversal-group theorems.

Corollary C.1 (Computed party-extension splittings). The realized projective party-permutation extension splits for both non-GRS $q = 11$ pencil classes, the $q = 11$ GRS evaluation-set representatives

$$\{0, 1, 2, 3, 4, 5\}, \{0, 1, 2, 3, 4, 6\}, \{0, 1, 2, 3, 5, 6\}, \{0, 1, 2, 3, 5, 9\},$$

the enhanced-symmetry $t = -1$ rows over $\mathbb{F}_{17}$ and $\mathbb{F}_{31}$, and the H_3 representatives $(q, \tau) = (11, 4), (19, 5), (29, 6), (31, 13)$. For every listed non-GRS or H_3 row, party parity acts on the fixed split torus T by inversion. Thus, after choosing an anchor leg and allowing party motion to transport the corresponding encoder view, a realized odd party permutation enlarges the possible local linear carrier exactly from T to

$$N(T) = \langle T, J \rangle, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I, \quad J \operatorname{diag}(a, a^{-1}) J^{-1} = \operatorname{diag}(a^{-1}, a).$$

Hence $N(T)/T \cong C_2$. The linear extension does not split in odd characteristic, although its projectivization does. If the party permutation fixes the input leg of a chosen encoder, this carrier acts as an actual logical enlargement; otherwise it compares two encoder views of the same six-leg tensor.

Proof. For each row, an exact finite computation reconstructs the fifteen four-support shortened planes, exhausts all 720 party permutations, and checks the propagated local symplectic blocks against the full CSS Lagrangian. After quotienting by the fixed-party group identified in Theorem 5.2, it is enough to test anchor block I on the GRS locus and anchor blocks I, J off it; this discards no candidate. In the order in which the six $q = 11$ rows are listed above, their party images are

$$S_5, \quad S_4, \quad V_4, \quad C_5, \quad S_3, \quad D_{12},$$

where D_{12} has order 12. The $q = 17$ and $q = 31$ enhanced rows have party images S_4 and S_5 , respectively, and each listed H_3 image is S_5 .

The certificate records the complete normalized factor set for each section. It also closes a complement of order $|\Pi|$ whose projection to Π is bijective, and verifies that the associated normalized cochain changes the factor set to the identity. Hence every listed extension splits. Conjugating the fixed torus by the stored lifts gives the sign action; any odd lift therefore induces the nontrivial class of $N(T)/T$. The complement is an involution in the projective party-extension group; its determinant-one anchor block is J , whose square is $-I$. $\square$

At $q = 11$, the two non-GRS pencil classes have party images S_5 and S_4 , their extensions split, and odd party motion inverts T . After choosing an anchor leg, the possible one-qudit affine carriers across transported encoder views therefore form $\mathbb{F}_q^2 \rtimes N(T)$. This is the logical group of a fixed encoder only for the subgroup fixing its input leg; general party motion compares different encoder views. The GRS fixed-party logical group and this anchored non-GRS carrier have orders 159720 and 2420, respectively.

Arithmetic symmetry jumps do not change the logical phase. The characteristic-17 tetrahedral specialization lies on the GRS boundary and retains the full $\mathrm{SL}_2(17)$ fixed-party image; its party image is S_4 , and the computed extension splits. The characteristic-31 A_5 specialization is non-GRS, so its fixed-party image is the split torus; its S_5 party image also splits, and odd motion enlarges the transported anchor-block carrier to $N(T)$.

References

- [ABO08] Dorit Aharonov and Michael Ben-Or. Fault-tolerant quantum computation with constant error rate. *SIAM Journal on Computing*, 38(4):1207–1282, 2008.
- [AJO14] Jonas T. Anderson and Tomas Jochym-O’Connor. Classification of transversal gates in qubit stabilizer codes, 2014. arXiv:1409.8320.
- [BL19] Simeon Ball and Michel Lavrauw. Arcs in finite projective spaces. *EMS Surveys in Mathematical Sciences*, 6(1–2):133–172, 2019.
- [CG26] Aranya Chakraborty and Daniel Gottesman. No-go theorem on fault tolerant gadgets for multiple logical qubits, 2026. arXiv:2602.13395.
- [Cob22] Arthur B. Coble. Associated sets of points. *Transactions of the American Mathematical Society*, 24(1):1–20, 1922.

- [CŚ06] Benoît Collins and Piotr Śniady. Integration with respect to the Haar measure on unitary, orthogonal and symplectic group. *Communications in Mathematical Physics*, 264(3):773–795, 2006.
- [DB25] Shival Dasu and Simon Burton. A classification of transversal Clifford gates for qubit stabilizer codes, 2025. arXiv:2507.10519.
- [DDM03] Jeroen Dehaene and Bart De Moor. The Clifford group, stabilizer states, and linear and quadratic operations over $\text{GF}(2)$. *Physical Review A*, 68(4):042318, 2003.
- [Dic01] Leonard Eugene Dickson. *Linear Groups: With an Exposition of the Galois Field Theory*. B. G. Teubner, Leipzig, 1901.
- [DO88] Igor Dolgachev and David Ortland. *Point Sets in Projective Spaces and Theta Functions*, volume 165 of *Astérisque*. Société Mathématique de France, 1988.
- [EK09] Bryan Eastin and Emanuel Knill. Restrictions on transversal encoded quantum gate sets. *Physical Review Letters*, 102:110502, 2009.
- [Fab23] Xander Faber. Finite p -irregular subgroups of $\text{PGL}_2(k)$. *La Matematica*, 2(2):479–522, 2023.
- [Gér76] Paul Gérardin. Three Weil representations associated to finite fields. *Bulletin of the American Mathematical Society*, 82(2):268–270, 1976.
- [GGB99] Markus Grassl, Willi Geiselmann, and Thomas Beth. Quantum Reed–Solomon codes. In *Applied Algebra, Algebraic Algorithms and Error-Correcting Codes*, volume 1719 of *Lecture Notes in Computer Science*, pages 231–244. Springer, 1999.
- [Got99] Daniel Gottesman. Fault-tolerant quantum computation with higher-dimensional systems. In *Quantum Computing and Quantum Communications*, volume 1509 of *Lecture Notes in Computer Science*, pages 302–313. Springer, 1999.
- [HDDM05] Erik Hostens, Jeroen Dehaene, and Bart De Moor. Stabilizer states and Clifford operations for systems of arbitrary dimensions, and modular arithmetic. *Physical Review A*, 71(4):042315, 2005.
- [HG20] Felix Huber and Markus Grassl. Quantum codes of maximal distance and highly entangled subspaces. *Quantum*, 4:284, 2020.
- [Hir98] J. W. P. Hirschfeld. *Projective Geometries over Finite Fields*. Oxford Mathematical Monographs. Clarendon Press, Oxford, 2 edition, 1998.
- [HMSV08] Benjamin Howard, John Millson, Andrew Snowden, and Ravi Vakil. A description of the outer automorphism of S_6 , and the invariants of six points in projective space. *Journal of Combinatorial Theory, Series A*, 115(7):1296–1303, 2008.
- [Hol26] Adam Holmes. Quantum logic codes: Complete transversal logical Clifford instruction sets for high-rate stabilizer quantum error correcting codes, 2026. arXiv:2606.13521.
- [HP03] W. Cary Huffman and Vera Pless. *Fundamentals of Error-Correcting Codes*. Cambridge University Press, 2003.

- [JCWY10] Zhengfeng Ji, Jianxin Chen, Zhaohui Wei, and Mingsheng Ying. The LU–LC conjecture is false. *Quantum Information & Computation*, 10(1–2):97–108, 2010.
- [KKKS06] Avanti Ketkar, Andreas Klappenecker, Santosh Kumar, and Pradeep Kiran Sarvepalli. Nonbinary stabilizer codes over finite fields. *IEEE Transactions on Information Theory*, 52(11):4892–4914, 2006.
- [PYHP15] Fernando Pastawski, Beni Yoshida, Daniel Harlow, and John Preskill. Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence. *Journal of High Energy Physics*, 2015(6):149, 2015.
- [Rai99] Eric M. Rains. Quantum codes of minimum distance two. *IEEE Transactions on Information Theory*, 45(1):266–271, 1999.
- [RGRA18] Zahra Raissi, Christian Gogolin, Arnau Riera, and Antonio Acín. Optimal quantum error correcting codes from absolutely maximally entangled states. *Journal of Physics A: Mathematical and Theoretical*, 51(7):075301, 2018.
- [RL25] Nithin Ramadas and Arul Lakshminarayan. Local unitary equivalence of absolutely maximally entangled states constructed from orthogonal arrays. *Journal of Physics A: Mathematical and Theoretical*, 58(12):125301, 2025.
- [Seg55] Beniamino Segre. Ovals in a finite projective plane. *Canadian Journal of Mathematics*, 7:414–416, 1955.
- [SKW⁺25] Hasan Sayginel, Stergios Koutsoumpas, Mark Webster, Abhishek Rajput, and Dan E. Browne. Fault-tolerant logical Clifford gates from code automorphisms, 2025. arXiv:2409.18175.
- [Tan26] Ian Tan. Transversal gates of the $((3, 3, 2))$ qutrit code and local symmetries of the absolutely maximally entangled state of four qutrits. *Discover Quantum Science*, 1(1), 2026.
- [TCT26] Theerapat Tansuwannont, Tim Chan, and Ryuji Takagi. Construction of the full logical Clifford group for high-rate quantum Reed–Muller codes using only transversal and fold-transversal gates, 2026. arXiv:2602.09788.
- [VdNDDM05a] Maarten Van den Nest, Jeroen Dehaene, and Bart De Moor. Finite set of invariants to characterize local Clifford equivalence of stabilizer states. *Physical Review A*, 72(1):014307, 2005.
- [VdNDDM05b] Maarten Van den Nest, Jeroen Dehaene, and Bart De Moor. Local unitary versus local Clifford equivalence of stabilizer states. *Physical Review A*, 71(6):062323, 2005.
- [Wir11] Klaus Wirthmüller. Automorphisms of stabilizer codes, 2011. arXiv:1102.5715.
- [ZL24] Canze Zhu and Qunying Liao. A class of double-twisted generalized Reed–Solomon codes. *Finite Fields and Their Applications*, 95:102395, 2024.
- [ZZ26] Yunlong Zhu and Chang-An Zhao. On iso-dual MDS codes from elliptic curves. *Finite Fields and Their Applications*, 109:102699, 2026.